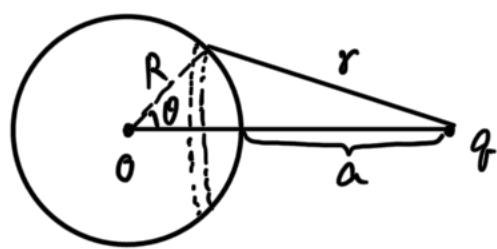


1. (1) 只需要論證單點電荷 q ，在球形區域產生的平均電勢 $\langle \varphi \rangle$

$$\text{滿足 } \langle \varphi \rangle = \frac{1}{4\pi R^2} \int_S \varphi_g dS = \varphi_{\text{球心}, q}$$



$$\varphi_g = \frac{q}{4\pi\epsilon_0 [R^2 + (R+a)^2 - 2R(R+a)\cos\theta]^{3/2}}$$

$$\begin{aligned} \langle \varphi_g \rangle &= \frac{1}{4\pi R^2} \cdot \frac{q}{4\pi\epsilon_0} \cdot \int_0^\pi \frac{2\pi R \sin\theta \cdot R d\theta}{[R^2 + (R+a)^2 - 2R(R+a)\cos\theta]^{3/2}} \\ &= \frac{1}{2R} \cdot \frac{q}{4\pi\epsilon_0} \int_0^\pi \frac{\sin\theta d\theta}{[1 + (1+\frac{a}{R})^2 - 2(1+\frac{a}{R})\cos\theta]^{3/2}} \\ &= \frac{1}{2(a+R)} \cdot \frac{q}{4\pi\epsilon_0} \int_0^\pi \frac{d[-(1+\frac{a}{R})\cos\theta]}{[1 + (1+\frac{a}{R})^2 - 2(1+\frac{a}{R})\cos\theta]^{3/2}} \\ &= \frac{1}{2(a+R)} \cdot \frac{q}{4\pi\epsilon_0} \int_{-(1+\frac{a}{R})}^{1+\frac{a}{R}} \frac{dx}{\sqrt{1 + (1+\frac{a}{R})^2 + 2x}} \\ &= \frac{1}{2(a+R)} \cdot \frac{q}{4\pi\epsilon_0} \cdot \left[\sqrt{1 + (1+\frac{a}{R})^2 + 2x} \right]_{-(1+\frac{a}{R})}^{1+\frac{a}{R}} \\ &= \frac{1}{2(a+R)} \cdot \frac{q}{4\pi\epsilon_0} \left[(2 + \frac{a}{R}) - (\frac{a}{R}) \right] \\ &= \frac{q}{4\pi\epsilon_0(a+R)} = \varphi_{\text{球心}, q} \end{aligned}$$

$$\begin{aligned} \langle \varphi \rangle &= \frac{1}{4\pi R^2} \int_S \varphi dS \\ &= \frac{1}{4\pi R^2} \int_S \sum_i \varphi_{q_i} dS \\ &= \sum_i \varphi_{\text{球心}, q_i} \\ &= \varphi_{\text{球心}} \end{aligned}$$

2. $\rho(\vec{r}) = -\epsilon_0 \nabla^2 \varphi$

$$\begin{aligned} &= -\epsilon_0 \nabla \cdot \left(\frac{q}{4\pi\epsilon_0} \cdot \frac{-be^{-br}r - e^{-br}}{r^2} \cdot \hat{r} \right) \\ &= \frac{q}{4\pi} \nabla \cdot \left(\frac{(br+1)e^{-br}}{r^2} \hat{r} \right) \\ &= \frac{q}{4\pi} \left[\nabla \cdot \left((br+1)e^{-br} \right) \cdot \frac{\hat{r}}{r^2} + (br+1)e^{-br} \nabla \cdot \left(\frac{\hat{r}}{r^2} \right) \right] \\ &= \frac{q}{4\pi} \left[(be^{-br} - b(br+1)e^{-br}) \hat{r} \cdot \frac{\hat{r}}{r^2} + 4\pi\delta(\vec{r})(br+1)e^{-br} \right] \\ &= \frac{q}{4\pi} \left[-\frac{b^2 e^{-br}}{r} + 4\pi\delta(\vec{r})(br+1)e^{-br} \right] \\ &= \frac{qe^{-br}}{4\pi} \left[-\frac{b^2}{r} + 4\pi\delta(\vec{r})(br+1) \right] \end{aligned}$$

3.

$$\begin{aligned} \vec{p} &= Qa\hat{x} + (-Q)(-a)\hat{x} + 2Q \cdot 2l\hat{y} \\ &= 2Q(a\hat{x} + 2l\hat{y}) \\ |\vec{p}| &= 2Q\sqrt{a^2 + (2l)^2} \end{aligned}$$

$$\begin{aligned} D_{ij} &= \int 3x_i x_j \rho(\vec{x}') dV' \\ &= 3 \int x_i x_j \rho_x \end{aligned}$$

$$D_{11} = 3 \times [a^3 \times (-Q) + a^3 \times Q + 0^3 \times 2Q] = 0$$

$$D_{22} = 3 \times [(2l)^3 \times 2Q] = 24l^3 Q$$

$$D_{33} = 0$$

$$D_{12} = D_{21} = 0$$

$$D_{13} = D_{31} = 0$$

$$D_{23} = D_{32} = 0$$

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 24l^3 Q & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

4. (1) $W_{r < a} = \oint_{|F| < a} \frac{1}{2} \vec{D} \cdot \vec{E} dV = \frac{\epsilon_0}{2} \oint_{|F| < a} E^2 dV$

$$\rho(\vec{r}) = \begin{cases} \frac{Q}{\frac{4}{3}\pi a^3}, & |\vec{r}| < a \\ 0, & |\vec{r}| \geq a \end{cases} \Rightarrow |\vec{E}(\vec{r})| = \begin{cases} \frac{Q \cdot (\frac{r}{a})^3}{4\pi\epsilon_0 r^2} = \frac{Qr}{4\pi\epsilon_0 a^3}, & |\vec{r}| < a \\ \frac{Q}{4\pi\epsilon_0 r^2}, & |\vec{r}| \geq a \end{cases} \Rightarrow \vec{E}(\vec{r}) = \begin{cases} \frac{Q}{4\pi\epsilon_0 a^3} \vec{r}, & |\vec{r}| < a \\ \frac{Q}{4\pi\epsilon_0 r^2} \vec{r}, & |\vec{r}| \geq a \end{cases}$$

$$\begin{aligned} W_{r < a} &= \frac{\epsilon_0}{2} \int_0^a \left(\frac{Q}{4\pi\epsilon_0 a^3} \right)^2 r^2 \cdot 4\pi r^2 dr \\ &= \frac{2\pi\epsilon_0}{5} \left(\frac{Q}{4\pi\epsilon_0 a^3} \right)^2 a^5 \\ &= \frac{Q^2}{40\pi\epsilon_0 a} \end{aligned}$$

$$\begin{aligned} (2) W_{r > a} &= \frac{\epsilon_0}{2} \int_a^\infty \left(\frac{Q}{4\pi\epsilon_0 r^2} \right)^2 \cdot 4\pi r^2 dr \\ &= \frac{\epsilon_0}{2} \cdot 4\pi \left(\frac{Q}{4\pi\epsilon_0} \right)^2 \int_a^\infty r^{-2} dr \\ &= \frac{Q^2}{8\pi\epsilon_0 a} \end{aligned}$$

$$\begin{aligned} (3) W &= W_{r < a} + W_{r > a} \\ &= \frac{3Q^2}{20\pi\epsilon_0 a} \end{aligned}$$

(2) 考慮球內單點電荷 q ，對球面的電位是 φ_g



$$\varphi_g = \frac{q}{4\pi\epsilon_0 [R^2 + (R-a)^2 - 2R(R-a)\cos\theta]^{3/2}}$$

$$\begin{aligned} \langle \varphi_g \rangle &= \frac{1}{4\pi R^2} \cdot \frac{q}{4\pi\epsilon_0} \cdot \int_0^\pi \frac{2\pi R \sin\theta \cdot R d\theta}{[R^2 + (R-a)^2 - 2R(R-a)\cos\theta]^{3/2}} \\ &= \frac{1}{2R} \cdot \frac{q}{4\pi\epsilon_0} \int_0^\pi \frac{\sin\theta d\theta}{[1 + (1-\frac{a}{R})^2 - 2(1-\frac{a}{R})\cos\theta]^{3/2}} \\ &= \frac{1}{2(R-a)} \cdot \frac{q}{4\pi\epsilon_0} \int_0^\pi \frac{d[-(1-\frac{a}{R})\cos\theta]}{[1 + (1-\frac{a}{R})^2 - 2(1-\frac{a}{R})\cos\theta]^{3/2}} \\ &= \frac{1}{2(R-a)} \cdot \frac{q}{4\pi\epsilon_0} \int_{-(1-\frac{a}{R})}^{1-\frac{a}{R}} \frac{dx}{\sqrt{1 + (1-\frac{a}{R})^2 + 2x}} \\ &= \frac{1}{2(R-a)} \cdot \frac{q}{4\pi\epsilon_0} \cdot \left[\sqrt{1 + (1-\frac{a}{R})^2 + 2x} \right]_{-(1-\frac{a}{R})}^{1-\frac{a}{R}} \\ &= \frac{1}{2(R-a)} \cdot \frac{q}{4\pi\epsilon_0} \left[(2 - \frac{a}{R}) - (\frac{a}{R}) \right] \\ &= \frac{q}{4\pi\epsilon_0 R} \end{aligned}$$

$$\begin{aligned} \langle \varphi \rangle &= \sum_i \langle \varphi_{q_i} \rangle \\ &= \sum_i \frac{q_i}{4\pi\epsilon_0 R} = \frac{Q}{4\pi\epsilon_0 R} \end{aligned}$$